

2026 TURNAROUND MECHANICAL DECOKE

Vacuum Charge Furnace — PR-170029

Job Report | USADebusk | Cenovus Lima Refinery — Lima, OH

PROJECT NO.	FACILITY	EXECUTION	PROJECT MANAGER
USA26046	Cenovus Lima Refinery	Sep 19 – 30, 2026	Jesse Utsey
WORK ORDER	SCOPE	HEATER	DURATION
82036776	TA Mechanical Decoke	PR-170029	12 days

195.5	52	97	8
OPERATING HOURS	STAND-BY HOURS	PIGS RUN	PASSES

Project Information

Customer Details

FACILITY	Cenovus Lima Refinery
ADDRESS	1150 S Metcalf St, Lima, OH 45804
WORK ORDER	82036776
CONTACT	Alfredo Trevino

Project Details

SCOPE	2026 turnaround mechanical decoke of the Vacuum Charge Furnace, with filtration and smart-pig support
EXECUTION	September 19 – 30, 2026 (12 days)
HEATER	PR-170029 Vacuum Charge Furnace
EQUIPMENT	(1) Trimax Pumper (Trimax 6) · (1) Support Unit · (1) Filter Press (Press 1) · Crew Truck

Crew Details

PROJECT MANAGER	Jesse Utsey
SHIFT LEAD	Day — Jesse Utsey; Night — Sam Mixon
DAYSHIFT	Danilo Ramirez, Blake Harman, Blake Baugh, Travis Trenholm, David Charles
NIGHTSHIFT	Tyler Garza, Joel Ruiz, Mikee Green

Project Duration

SCOPE	UNIT	RIG	PIG	SMART PIG	SUBTOTAL
PR-170029 Vacuum Charge Furnace	Trimax 6	58.5	63	74	195.5

Stand-By Summary

Stand-by is on-site time during which the pumping units were not actively pigging. Cause is recorded from the daily service receipts.

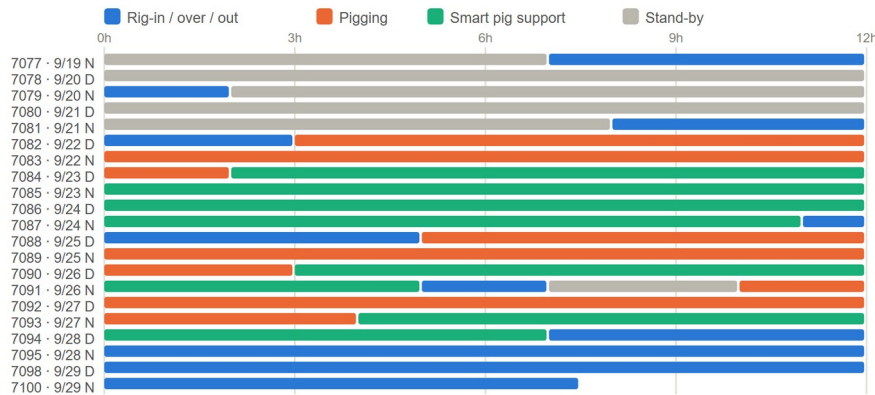
DATES	HOURS	CAUSE
Sep 19 – 21 (7077–7081)	49	Awaiting installation of the 12" radiant-outlet launchers (Vacuum)
Sep 26 (7091)	3	Awaiting relocation of the 12" launchers to passes 5, 7, 8 (Vacuum)
TOTAL	52	

Heater total: 195.5 operating hours plus 52 stand-by hours — 247.5 pumping-unit hours.

PR-170029 — Vacuum Charge Furnace

Number of Passes	8, pigged individually
Total Footage	7,054 ft (882 ft per pass)
Convection Tube ID	4.000"
Radiant Tube ID	3.826" · 5.761" · 7.813" · 9.750" · 11.750" (4" to 12")
Metallurgy	A312 TP317/317L
Return Bends	180° seamless
Smart Pigging	Yes — Intero

Decoking Analysis: Passes 4 and 5 carried the heaviest fouling, pass 5 the most. In pass 5, fouling was heavy in the 12" section, moderate in the 10" and 4" sections, and light in the 6" and 8" sections. In pass 4, it was moderate in the 12", light to moderate in the 10", and light in the 8", 6" and 4" sections. The remaining six passes showed light fouling throughout. The external 6" convection inlet piping, between the control valve and the 4" convection inlet tube, was heavily fouled on all eight passes, enough to keep 4" pigs from entering the convection section, and was cleaned first on each pass. Average pigging time was approximately 21 hours per pass.



PR-170029 - task hours by shift, from the daily service receipts.

Pigs Used

SIZE — QTY	SIZE — QTY	SIZE — QTY	SIZE — QTY
3.75" — 4	4.125" — 15	6.25" — 6	10" — 20
3.875" — 1	4.25" — 4	8" — 7	12" — 6
4" — 22	6" — 10	8.25" — 2	
TOTAL — 97			

Flow Tests

Pass 1

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	295	220	75
650	260	185	75
600	225	165	60

Pass 2

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	275	225	50
650	235	195	40
600	205	170	35

Pass 3

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	332	282	50
650	290	250	40
600	255	220	35

Pass 4

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	280	210	70
650	235	188	47
600	210	160	50

Pass 5

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	270	220	50
650	235	190	45
600	202	160	42

Pass 6

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	285	225	60
650	235	195	40
600	210	165	45

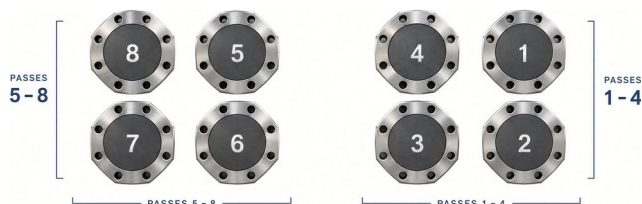
Pass 7

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	260	220	40
650	225	185	40
600	195	160	35

Pass 8

GPM	BEFORE PSI	AFTER PSI	Δ PSI
700	320	280	40
650	280	250	30
600	240	220	20

Images



Radiant outlet labels and orientation, viewed facing the outlets. Each outlet connects to the convection inlet of the same pass number.

Top: coke removed from the 12" tube, pass 5. Bottom: deposits removed from the external 6" convection inlet piping, collected from all eight passes.

Project Summary

USADebusk performed the mechanical decoke of the Vacuum Charge Furnace (PR-170029) for Cenovus at the Lima Refinery during the 2026 turnaround, September 19 – 30, on Trimax 6 with filtration. The external 6" convection inlet piping was heavily fouled on all eight passes and was cleaned first on each pass, before pigging inside the heater. All eight passes were smart-pigged by Intero after cleaning. Of the heater's 247.5 pumping-unit hours, 52 were stand-by, waiting on the 12" radiant-outlet launchers to be installed and later relocated.

Project Close

The Vacuum Charge Furnace was cleaned and returned to Cenovus, and all equipment was demobilized from the unit on September 30, 2026.

USADebusk thanks the Cenovus Lima team for their coordination and support throughout the turnaround, and we look forward to continuing to serve your decoking and pigging needs.

Jesse Utsey | Project Manager, USADebusk

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